Advanced Machine Learning, ETH Zurich

U-Nets

Carlos Cotrini

Autumn semester 2026

- A typical architecture for image segmentation.
- One can also treat a U-net as some sort of image autoencoder
- Image from original paper: https://arxiv.org/abs/1505.04597

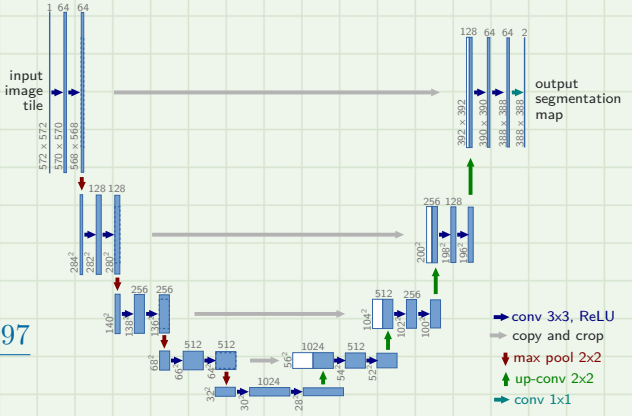
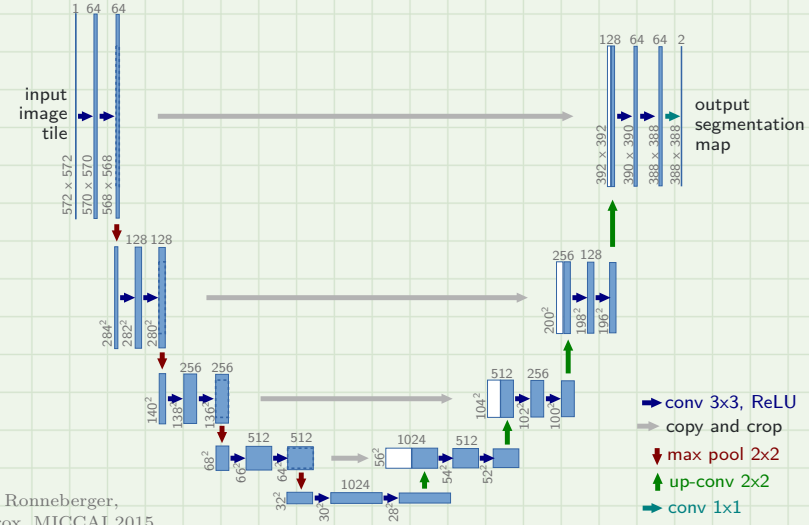


Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015



After Fig. 1 of Ronneberger, Fischer and Brox, MICCAI 2015